

MLLM-guided Training-free Spherical Panorama Generation from a Single Image

Yuki Katayama
Hokkaido University
Sapporo, Japan
katayama@lmd.ist.hokudai.ac.jp

Ren Togo
Hokkaido University
Sapporo, Japan
togo@lmd.ist.hokudai.ac.jp

Keisuke Maeda
Hokkaido University
Sapporo, Japan
maeda@lmd.ist.hokudai.ac.jp

Takahiro Ogawa
Hokkaido University
Sapporo, Japan
ogawa@lmd.ist.hokudai.ac.jp

Miki Haseyama
Hokkaido University
Sapporo, Japan
mhaseyama@lmd.ist.hokudai.ac.jp

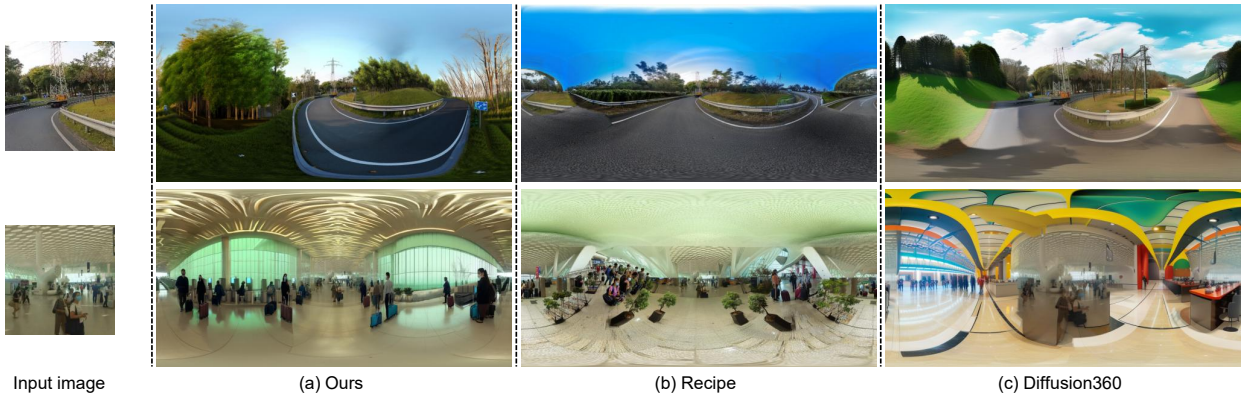


Figure 1: Visual comparison of panorama generation for outdoor (top) and indoor (bottom) scenes.

Abstract

We propose a novel training-free pipeline for generating a spherical scene from a single image. Prior methods suffer from two issues. Specifically, visual inconsistency appears as style drift from the input image. Another issue is structural distortion that degrades the global scene geometry. We address these issues by leveraging a Multi-modal Large Language Model to generate direction-conditioned prompts autoregressively and by employing a short multi-step outpainting process with large overlaps. Our method maintains cross-view consistency and preserves global scene geometry across all directions without additional training.

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1 Introduction

The generation of immersive panoramic scenes from a single image has attracted attention, especially for VR applications [Schwarz et al. 2025]. While fine-tuned methods like Diffusion360 [Feng et al. 2023] achieve visual consistency, they are inherently dataset-dependent and often suffer from style drift relative to the input image. To avoid this, training-free approaches such as Recipe [Schwarz et al. 2025] have been proposed. However, these methods, which may involve large viewpoint transformations, frequently induce structural distortions in Fig. 1. Moreover, both fine-tuned and training-free baselines rely on a single global prompt, leading to weakly constrained direction-wise semantics and reduced cross-view coherence.

To address these issues, we propose a novel training-free pipeline based on sequential outpainting for spherical panorama generation. A key novelty is introducing a Multi-modal Large Language Model (MLLM) to strengthen cross-view semantic coupling. We leverage an MLLM to extract global scene context from the input view and to autoregressively generate directional local context, conditioned on the same global context across views. (1) To mitigate visual inconsistency, we design a short multi-step outpainting process with sufficient overlap under a shared global context. (2) To reduce structural distortion, we keep pitch fixed during horizontal outpainting and leverage direction-wise local context. Our approach enables the generation of spatially coherent and input-faithful spherical panoramas from a single image without domain-specific training.

2 Method

Our method autoregressively generates directional prompts by using MLLM and follows them with a short multi-step sequential outpainting process for panorama generation.

2.1 MLLM-guided Prompt Generation

To strengthen cross-view semantic coupling, we use an MLLM to extract global scene context and autoregressively generate direction-wise prompts conditioned on it. We define the input image as I_f and assume it represents the front view of the scene. We denote by $\text{MLLM}_{\text{atmos}}$ the MLLM configured to extract from I_f the panorama-wide scene atmosphere (place, weather, time, style) as global context. Let g be the global atmosphere prompt:

$$g = \text{MLLM}_{\text{atmos}}(I_f).$$

Furthermore, we define $\text{MLLM}_{\text{scene}}$ as the MLLM that produces direction-wise local prompts (view-specific geometry, layout, and content) conditioned on g . For horizontal views, it generates these prompts autoregressively so semantics from earlier views propagate to later ones. Accordingly, we generate directional prompts P_{d_k} autoregressively as

$$P_{d_k} = \begin{cases} \text{MLLM}_{\text{scene}}(I_f, g, d_k), & k \in \{1, 5, 6\}, \\ \text{MLLM}_{\text{scene}}(I_f, g, d_k, \{P_{d_1}, \dots, P_{d_{k-1}}\}), & 1 < k \leq 4, \end{cases}$$

where the order $(d_1, \dots, d_6) = (\text{front}, \text{right}, \text{back}, \text{left}, \text{top}, \text{bottom})$. To share the global context, each P_{d_k} encodes only the elements of g pertinent to that view.

2.2 A Short Multi-step Sequential Outpainting

After obtaining the directional prompts P_{d_k} , we estimate the input FoV and set it as the front view (yaw 0° , pitch 0°), following Recipe [Schwarz et al. 2025]. We then outpaint each direction-conditioned view I_{d_k} using its corresponding prompt P_{d_k} . For horizontal sweeps, the pitch is fixed at 0° to avoid stretching the polar regions. In practice, we use an off-the-shelf text-to-image model [Zhang et al. 2023]. To ensure visual consistency, each outpainting step is conditioned on global context and uses sufficient overlap to propagate appearance across views. For the horizontal region, we set the FoV to 120° and perform 4-step outpainting; if the estimated FoV of I_f is below 80° , we split the front view into two. For the vertical region (top and bottom), we apply an FoV of 90° and use P_{top} and P_{bottom} as the horizontal sweeps already establish sufficient overlap.

3 Experiments

We compared our method with Diffusion360 [Feng et al. 2023] (a publicly available fine-tuned baseline) and Recipe [Schwarz et al. 2025] (a training-free baseline). As Recipe’s code is unavailable, we re-implemented it following its description, using identical image generators, resolution, and parameters.

We use input images from WonderJourney [Yu et al. 2024] and 40° front-view crops from Flickr360 [Cao et al. 2023] and Laval Indoor [Gardner et al. 2017], covering both outdoor and indoor scenes. We evaluate 50 images per dataset. Following Recipe, we adopt BRISQUE [Mittal et al. 2012] and Q-Align [Wu et al. 2024], both reported to correlate with human perception. As these metrics do not fully capture visual consistency and structural coherence,

Table 1: Quantitative comparison on WJ-Input, Flickr360 and Laval Indoor. Metrics are averaged over six perspective views per panorama (80° FoV, 60° yaw steps).

Method	WJ-Input		Flickr360		Laval Indoor	
	BRISQUE↓	Q-Align↑	BRISQUE↓	Q-Align↑	BRISQUE↓	Q-Align↑
Diffusion360	60	2.5	62	1.6	74	1.6
Recipe	44	2.6	44	1.7	51	1.6
Ours	43	2.9	43	2.0	53	1.7

we conducted a qualitative evaluation to verify our method’s effectiveness in preserving these aspects.

Table 1 shows the quantitative results. Our method achieves the best BRISQUE on WonderJourney Input (WJ-Input) and Flickr360, and the highest Q-Align across all datasets, indicating superior perceptual quality. Qualitatively, our method achieves structural coherence and visual consistency across all directions, which Diffusion360 and Recipe fail to maintain, as shown in Fig. 1.

4 Conclusion

We have proposed a training-free pipeline for spherical panorama generation that combines MLLM-guided autoregressive direction-conditioned prompts with a short multi-step outpainting process. Our pipeline ensures visual consistency and structural coherence across all directions without requiring additional training. On the other hand, our method can still produce subtle boundaries within the generated regions. We plan to introduce a feedback mechanism to further enforce structural coherence under the training-free setting. Our proposed method opens new possibilities for applications in tourism, entertainment, and immersive VR experiences, and is easily deployable and immediately usable.

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